

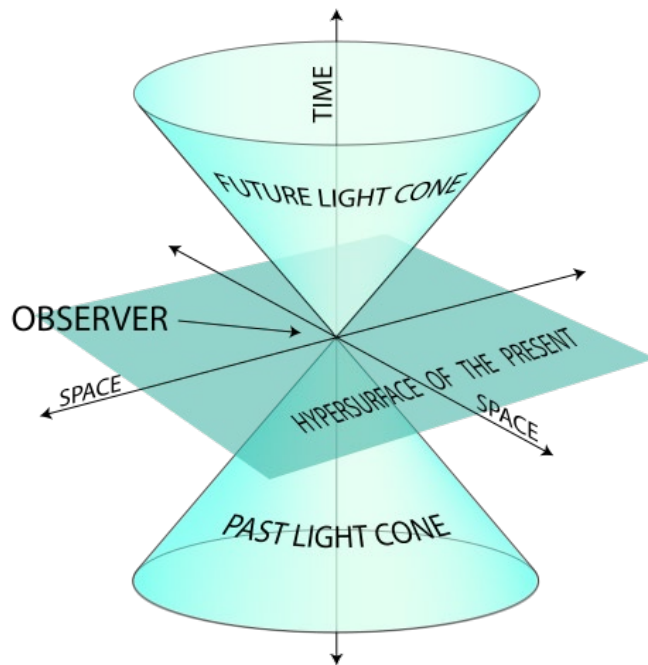
ELEC3115 Electromagnetic Engineering

Part B – High-frequency electromagnetics



Special relativity

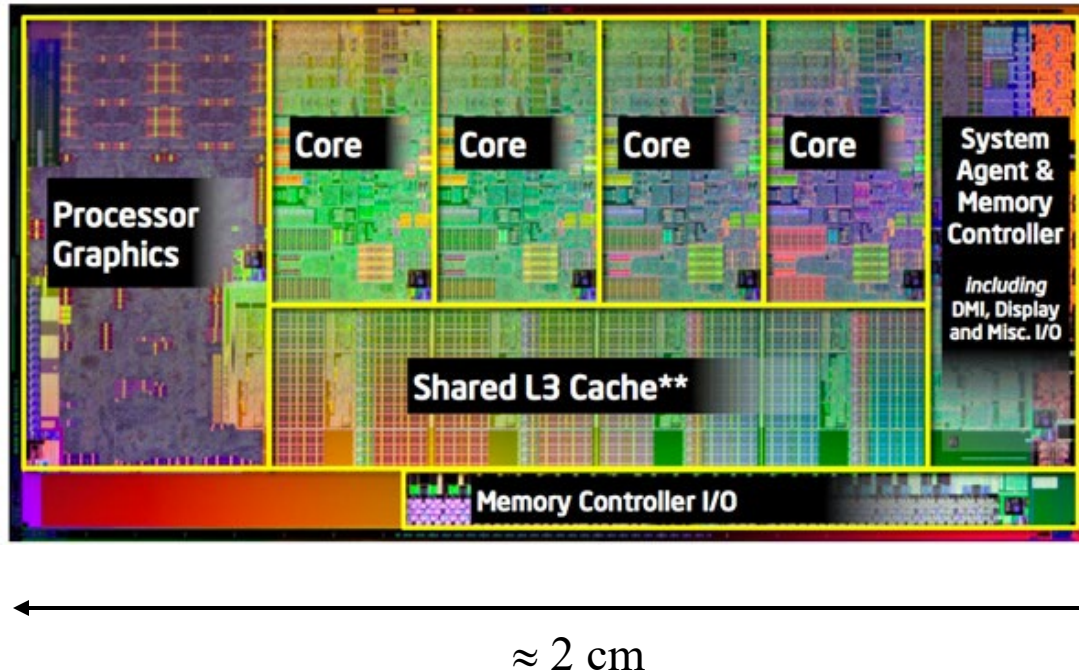
Information cannot travel faster than the speed of light



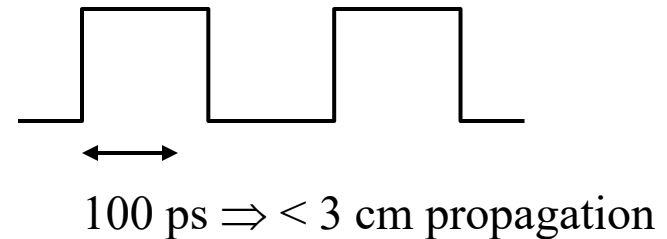
$$c \approx 3 \times 10^8 \text{ m/s (in vacuum)}$$

Light (and therefore **information**)
travels at most 30 cm
in 1 nanosecond

Important at high frequency!

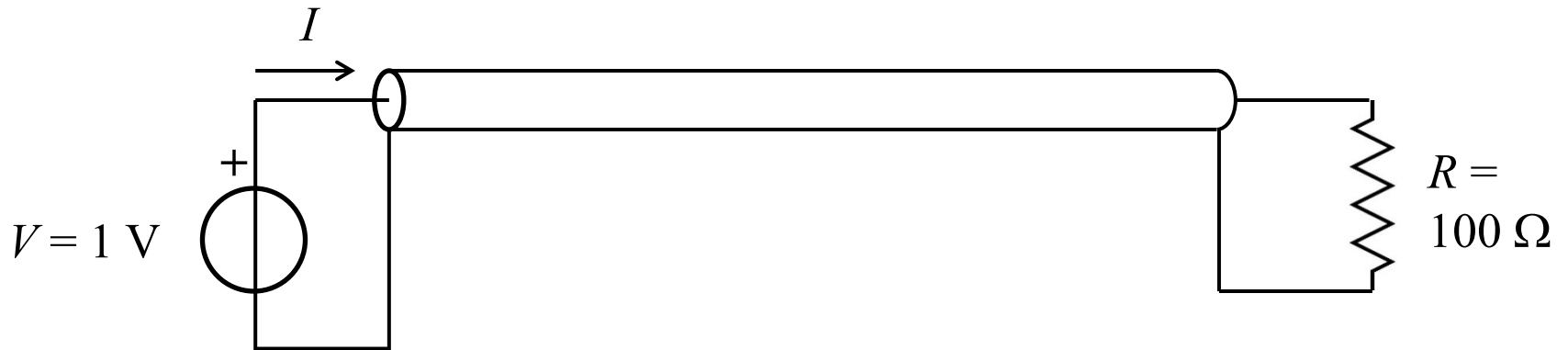


Clock speed $\approx 3\text{ GHz}$



Modern CPUs are multi-core, because information cannot travel fast enough (compared to the clock period) from one end of the chip to the other! So the chip needs to be sliced up into more compact “cores”.

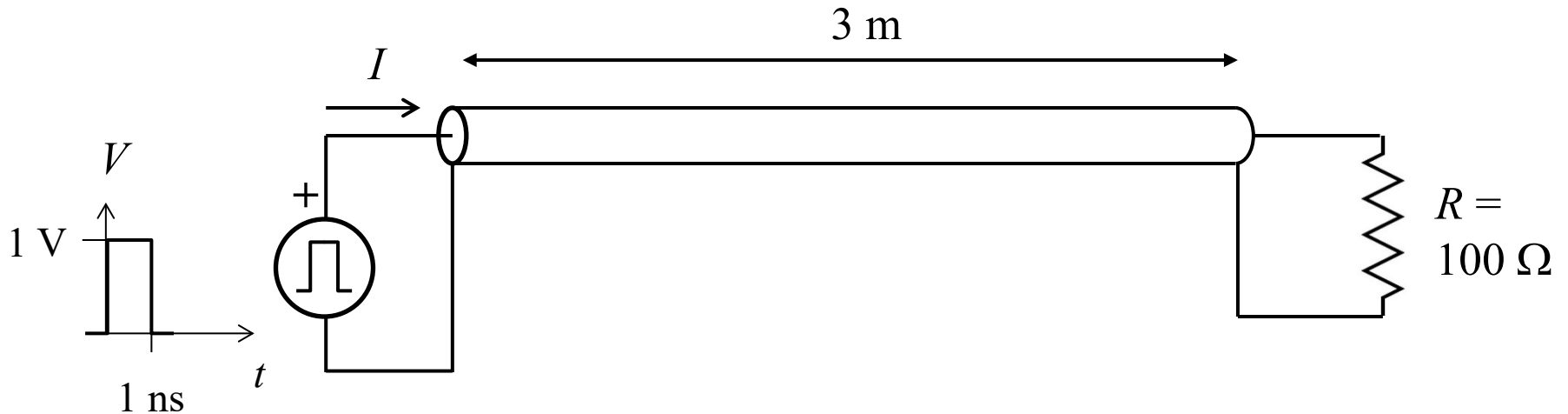
A simple example



How much current flows in the cable?

$$I = \frac{V}{R} = 10 \text{ mA}$$

A tricky example



How much is the current now?

$$I \neq \frac{V}{R}$$

the pulse generator cannot “know” the resistance at the other end of a 3 m long cable, if it sends out a pulse that only lasts for 1 ns !!

Electrical length

Consider a system whose purpose is to handle or transmit signals that vary significantly over a timescale τ , and has a physical size L .

In the time τ , light can propagate at most to a distance $c\tau$.

The electrical length of the system is

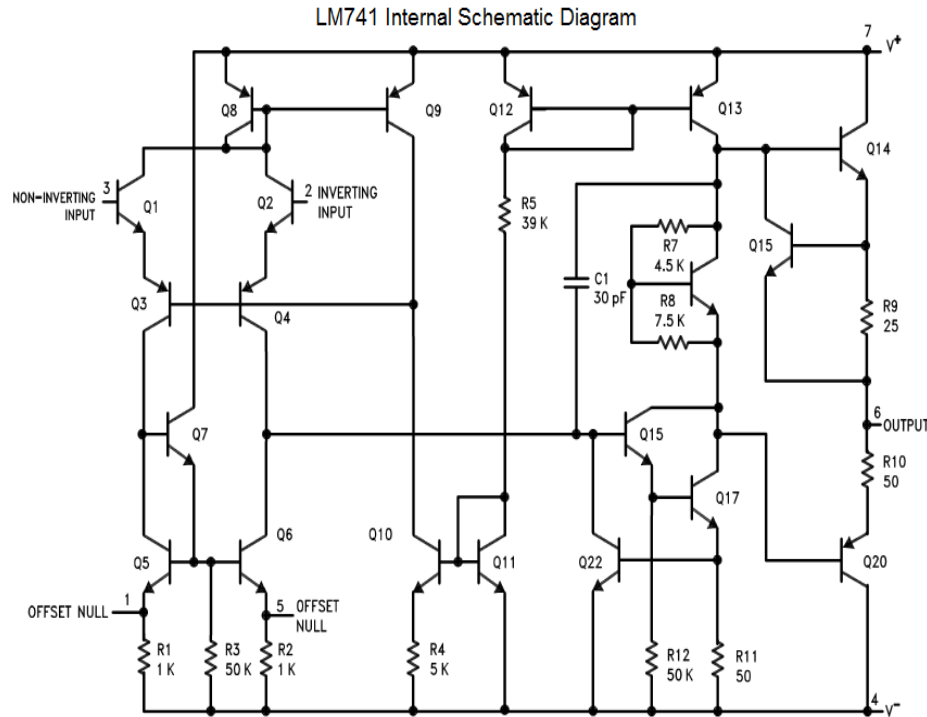
$$L_e = \frac{L}{c\tau}$$

In the example before, $L_e = 10$.

Note: In general, use $L_e = \frac{L}{v\tau}$, where v is the signal propagation speed, which may not be equal to c , the speed of light.

The aim of Part B of this course is to understand the behaviour of systems with $L_e \gg 1$

Lumped elements



When we analyse this circuit, we assume that each circuit element has “concentrated” (**lumped**) properties.

For example, a resistor has only the property of having a resistance between its two terminals. Its behaviour in a circuit is such that, when passing a current I , it imposes a voltage drop $V = RI$ across its terminals.

This is only true if the physical size of the resistor is small compared to the distance travelled by light within the timescale of relevance to the problem!

Also, we assume that nodes connected by a perfect wire have the same voltage, e.g. the collector of Q6 and the base of Q15. Implicitly, this means we assume that Q6 and Q15 are “within the same light cone” for the signal.

Distributed elements



In a coaxial cable, you can measure:

- The resistance of each conductor from one end to the other
- The inductance of each conductor from one end to the other
- The capacitance between the inner and outer conductors
- The conductance between the inner and outer conductors

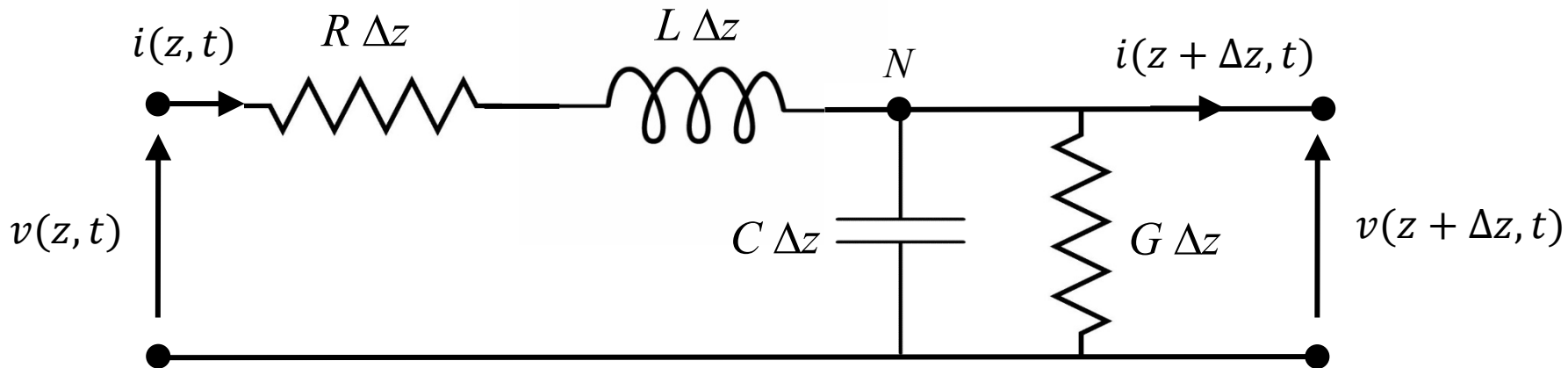
These parameters are not lumped! They are **distributed** all along the length of the cable.

If you double the length, each parameter doubles.

NEW: How do we analyse a circuit with distributed elements?

Lumped elements approximation

Consider a very short piece of cable, of length Δz . We can approximate it with lumped elements



R = series resistance per unit length (Ω / m)
 L = series inductance per unit length (H / m)
 C = shunt capacitance per unit length (F / m)
 G = shunt conductance per unit length (S / m)

Note: R and L are the combined resistance and inductance of the two conductors

Note: in a perfect cable $R = G = 0$
However, L and C are never zero

Typically:
 $C \approx 100 \text{ pF} / \text{m}$
 $L \approx 250 \text{ nH} / \text{m}$

Telegrapher's equations

Kirchoff's voltage law:

$$v(z + \Delta z, t) = v(z, t) - R\Delta z i(z, t) - L\Delta z \frac{\partial i(z, t)}{\partial t}$$
$$-\frac{v(z + \Delta z, t) - v(z, t)}{\Delta z} = Ri(z, t) + L \frac{\partial i(z, t)}{\partial t}$$

Kirchoff's current law at node N :

$$i(z + \Delta z, t) = i(z, t) - G\Delta z v(z + \Delta z, t) - C\Delta z \frac{\partial v(z + \Delta z, t)}{\partial t}$$
$$-\frac{i(z + \Delta z, t) - i(z, t)}{\Delta z} = Gv(z + \Delta z, t) + C \frac{\partial v(z + \Delta z, t)}{\partial t}$$

Telegrapher's equations

Taking the limit $\Delta z \rightarrow 0$:

$$-\frac{\partial v(z, t)}{\partial z} = Ri(z, t) + L \frac{\partial i(z, t)}{\partial t} \quad (1a)$$

$$-\frac{\partial i(z, t)}{\partial z} = Gv(z, t) + C \frac{\partial v(z, t)}{\partial t} \quad (1b)$$

For sinusoidal signals at frequency ω (phasors), use Fourier transform:

$$-\frac{dV(z)}{dz} = (R + j\omega L)I(z) \quad (2a)$$

$$-\frac{dI(z)}{dz} = (G + j\omega C)V(z) \quad (2b)$$

These are called **Telegrapher's equations**

Wave equations

Take d/dz of (2a), then substitute (2b) in it:

$$\frac{d^2 V(z)}{dz^2} = (R + j\omega L)(G + j\omega C)V(z) = \gamma^2 V(z) \quad (3a)$$

Same for (2b):

$$\frac{d^2 I(z)}{dz^2} = (R + j\omega L)(G + j\omega C)I(z) = \gamma^2 I(z) \quad (3b)$$

Where we have defined the **propagation constant**:

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta \quad (4)$$

In a lossless cable, $\gamma = j\beta = j\omega\sqrt{LC}$

Eqns. (3) are **wave equations**.

Propagating waves

Eqns. (3) are solved by waves propagating in opposite directions:

$$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{+\gamma z} \quad (5a)$$

$$I(z) = I_0^+ e^{-\gamma z} + I_0^- e^{+\gamma z} \quad (5b)$$

V_0^+ and I_0^+ are the amplitudes of the voltage and current associated with a wave that propagates along the positive z direction (“left to right”)

V_0^- and I_0^- are the amplitudes of the voltage and current associated with a wave that propagates along the negative z direction (“right to left”)

Notes:

- V_0^+ and V_0^- depend on the initial and boundary conditions.
- $V_0^{+/-}$ and $I_0^{+/-}$ are not independent of each other!

Propagating waves

Wave propagation is best understood in the time domain.

Consider for simplicity a “perfect” (lossless) cable, with $R = G = 0$. Then,

$$\gamma = j\beta = j\omega\sqrt{LC} \quad (\text{recall Eq. (4)})$$

Consider a wave propagating left to right ($V_0^- = 0$):

$$v(z, t) = \text{Re}\{V(z)e^{j\omega t}\} = V_0^+ \text{Re}\{e^{j\omega t} e^{-j\beta z}\} \quad (6)$$

At $z = 0, t = 0$ we obviously have $v(0,0) = V_0^+$

However, we also have V_0^+ at every place and time where $\omega t = \beta z$, because $e^{j\omega t} e^{-j\beta z} = 1$ there and then.

$$v\left(\frac{\omega t}{\beta}, t\right) = V_0^+$$

Phase velocity

We deduce that Eqn. (5) describes the voltage in space and time associated with a wave that propagates with velocity

$$v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} \quad (7)$$

v_p is called **phase velocity**. You can think of it as the speed at which the peak of the wave propagates in space.

If $C = 100$ pF/m and $L = 250$ nH/m, then

$$v_p = \frac{1}{\sqrt{10^{-10} \times 2.5 \times 10^{-7}}} = 2 \times 10^8 \text{ m/s}$$

i.e. 2/3 of the speed of light in vacuum.

Wavelength

For a wave with angular frequency ω , i.e. period $T = \frac{2\pi}{\omega}$, we have

$$v\left(\frac{\omega T}{\beta}, T\right) = V_0^+ \text{ and } v(0, T) = V_0^+$$

$$\text{because } z = \frac{\omega T}{\beta} = \frac{2\pi}{\beta} \Rightarrow e^{-j\beta z} = e^{-j2\pi} = e^{-j0} = 1$$

After a time T equal to the period (in time) of the wave, the wave has moved a distance

$$\lambda = \frac{2\pi}{\beta} = T v_p = \frac{v_p}{f} \quad (8)$$

λ is the **wavelength**. It represent the period (in space) of the wave.

Characteristic impedance

Calculate (2a) using (5a):

$$\frac{dV(z)}{dz} = -(R + j\omega L)I(z) = -\gamma V_0^+ e^{-\gamma z} + \gamma V_0^- e^{+\gamma z}$$

$$I(z) = \frac{\gamma}{R + j\omega L} (V_0^+ e^{-\gamma z} - V_0^- e^{+\gamma z}) \quad (9)$$

Comparing (9) with (5b) we deduce that:

$$I_0^+ = V_0^+ \frac{\gamma}{R + j\omega L}$$

$$I_0^- = -V_0^- \frac{\gamma}{R + j\omega L}$$

Characteristic impedance

Therefore:

$$\frac{V_0^+}{I_0^+} = \frac{R + j\omega L}{\gamma} = \frac{R + j\omega L}{\sqrt{(R + j\omega L)(G + j\omega C)}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

We define the **characteristic impedance** of the cable as:

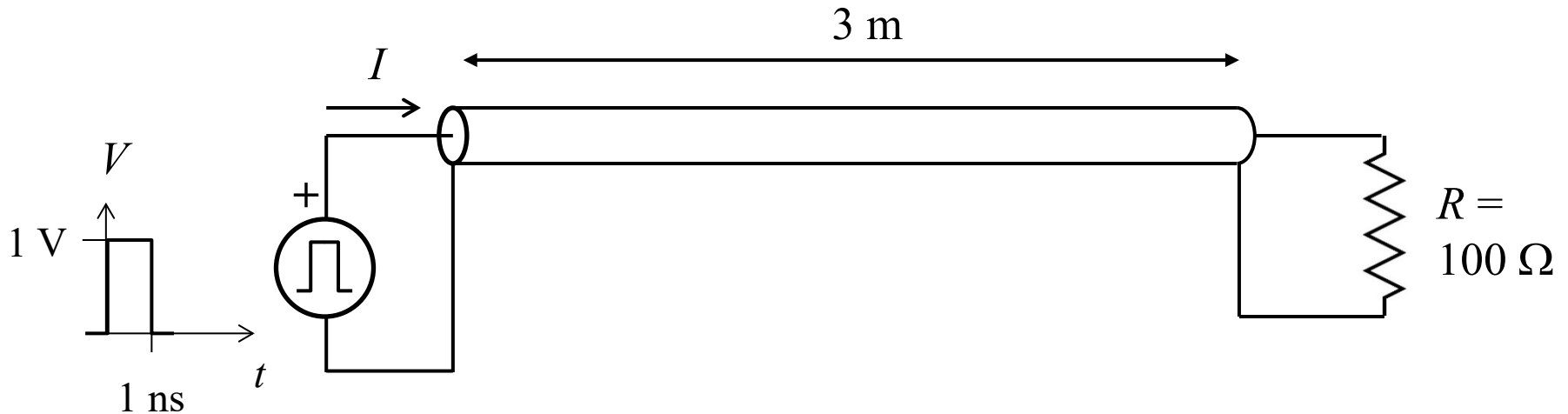
$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad (10)$$

The characteristic impedance is the ratio of the amplitudes of voltage and current **in the propagating wave**.

If the cable is lossless ($R = G = 0$) and $C = 100$ pF/m, $L = 250$ nH/m, then

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{2.5 \times 10^{-7}}{10^{-10}}} = 50 \, \Omega$$

The answer to the earlier question



The current is:

$$I = \frac{V}{Z_0}$$

If $Z_0 = 50\ \Omega$, then $I = 20\text{ mA}$.

This I is what we called I_0^+ before. It is completely independent of what value of resistance is attached at the other end of that cable!

Voltage and current on a line

Recall Eqns. (9) and (10):

$$I(z) = \frac{\gamma}{R + j\omega L} (V_0^+ e^{-\gamma z} - V_0^- e^{+\gamma z}) \quad (9)$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad (10)$$

Plug these into (5b): $I(z) = I_0^+ e^{-\gamma z} + I_0^- e^{+\gamma z}$:

$$I(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} - \frac{V_0^-}{Z_0} e^{+\gamma z} \quad (11)$$

Notice that $Z_0 = \frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-}$

The minus sign in the front of the expressions for the backwards-propagating waves is because we define the currents as positive going left to right.

Lossless transmission line

Recall Eq. (4)

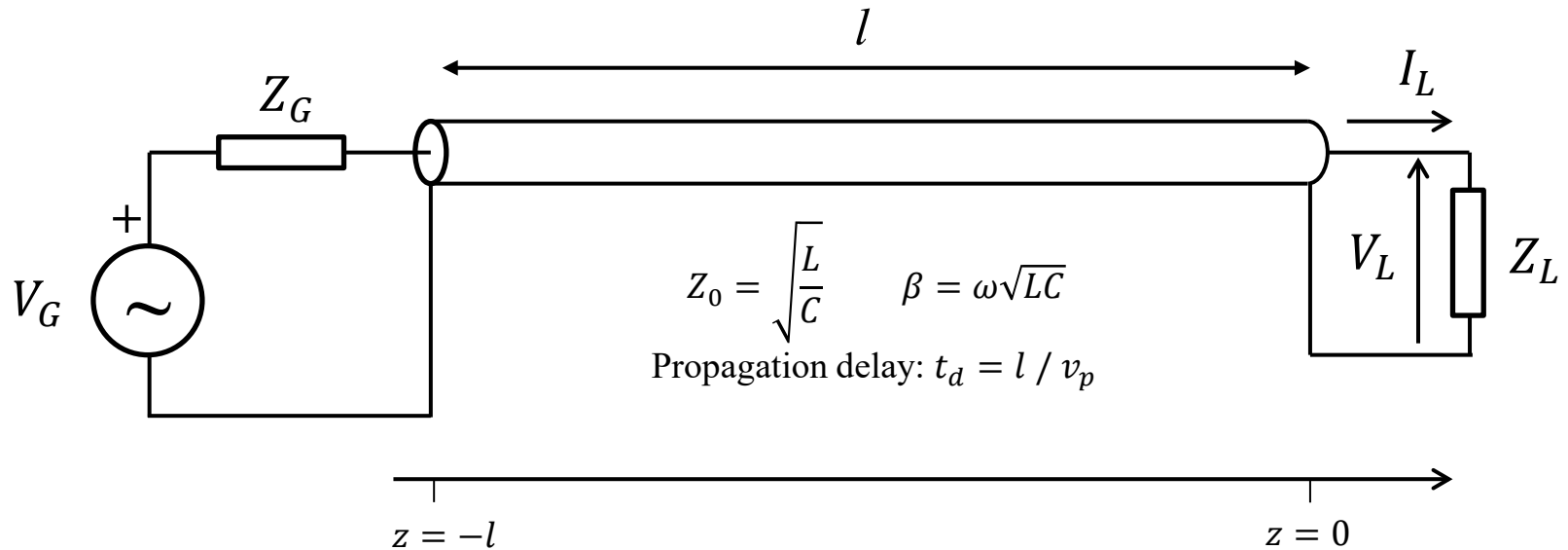
$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta$$

If there is no resistive dissipation along the cable, and no leakage current between the two conductors, i.e. $R = G = 0$, then:

$$\gamma = j\beta$$
$$Z_0 = \sqrt{\frac{L}{C}}$$

Attention! This is the opposite of what you are used to in low-frequency circuit analysis! Here, the characteristic impedance is **purely real** when you have **no resistive components**!

Terminated lossless line



Voltage and current on the line:

$$V(z) = V_0^+ e^{-j\beta z} + V_0^- e^{+j\beta z} \quad (12a)$$

$$I(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{+j\beta z} \quad (12b)$$

Boundary conditions

At the load ($z = 0$), from Eqns. (12):

$$V_L = V(0) = V_0^+ + V_0^- \quad (13a)$$

$$I_L = I(0) = I_0^+ + I_0^- = \frac{V_0^+ - V_0^-}{Z_0} \quad (13b)$$

The load impedance imposes a boundary condition:

$$\boxed{\frac{V_L}{I_L} = Z_L}$$

Using Eqns. (13):

$$Z_L = Z_0 \frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} \quad (14)$$

Solving for V_0^- :

$$Z_L(V_0^+ - V_0^-) = Z_0(V_0^+ + V_0^-)$$

$$V_0^- = \frac{Z_L - Z_0}{Z_L + Z_0} V_0^+ \quad (15)$$

Voltage reflection coefficient

From Eq. (15) we can calculate the ratio between the voltages of the reflected wave and the incident wave:

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} \quad (16)$$

Γ is the **voltage reflection coefficient**.

It is often expressed in terms of the normalized load impedance $z_L = \frac{Z_L}{Z_0}$:

$$\Gamma = \frac{z_L - 1}{z_L + 1} \quad (17)$$

We can rewrite Eqns. (12) as:

$$V(z) = V_0^+ (e^{-j\beta z} + \Gamma e^{+j\beta z}) \quad (18a)$$

$$I(z) = \frac{V_0^+}{Z_0} (e^{-j\beta z} - \Gamma e^{+j\beta z}) \quad (18b)$$

Open, shorted, and matched line

From Eq. (16), we find some useful limiting cases:

- $Z_L = \infty \Rightarrow \Gamma = 1$ **open line:** the incident voltage is fully reflected.
- $Z_L = 0 \Rightarrow \Gamma = -1$ **short-circuited line:** the incident voltage is fully reflected, but with opposite sign.
- $Z_L = Z_0 \Rightarrow \Gamma = 0$ **matched line:** the load impedance matches the cable (characteristic) impedance. No reflections occur.

Note: in a lossless line, Z_0 is a real number, but the load impedance might be complex, so in general Γ is complex too.

$$\Gamma = |\Gamma|e^{j\theta}$$

$$0 \leq |\Gamma| \leq 1$$

θ is called **phase angle**

Propagation in the time domain

Telegrapher's equations (1) for lossless line, in the time domain:

$$\begin{aligned} -\frac{\partial v(z, t)}{\partial z} &= L \frac{\partial i(z, t)}{\partial t} \\ -\frac{\partial i(z, t)}{\partial z} &= C \frac{\partial v(z, t)}{\partial t} \end{aligned}$$

They yield a wave equation:

$$\frac{\partial^2 v(z, t)}{\partial z^2} = LC \frac{\partial^2 v(z, t)}{\partial t^2}$$

With general solution:

$$v(z, t) = v^+ \left(t - \frac{z}{v_p} \right) + v^- \left(t + \frac{z}{v_p} \right)$$

Where $v_p = \frac{1}{\sqrt{LC}}$

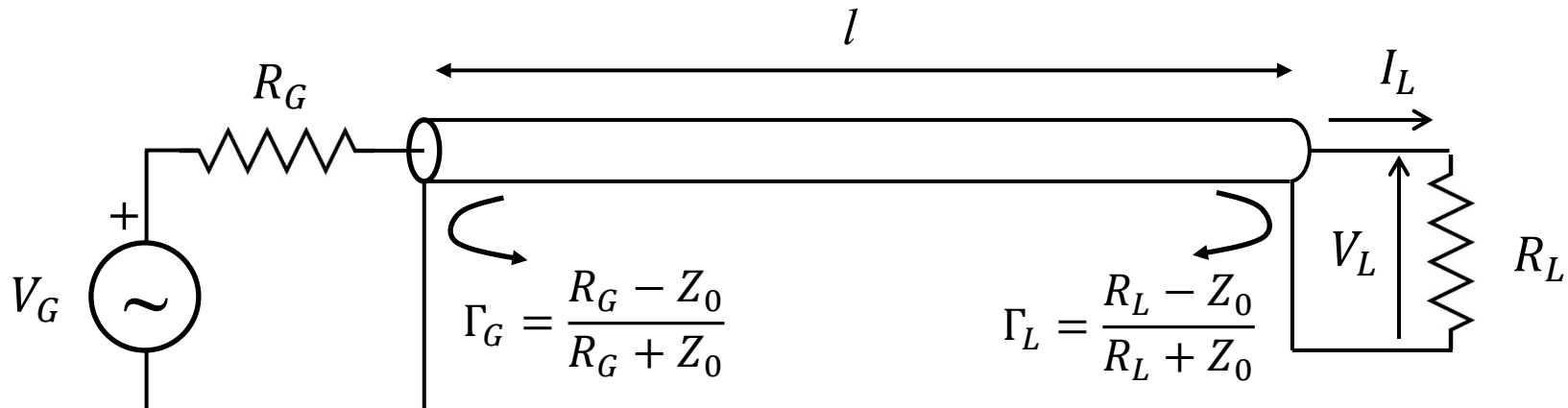
Reflections in the time domain

For a lossless line, Γ is frequency-independent. If the load impedance is purely real (resistive), then Γ is also real. Under these assumptions, the relation

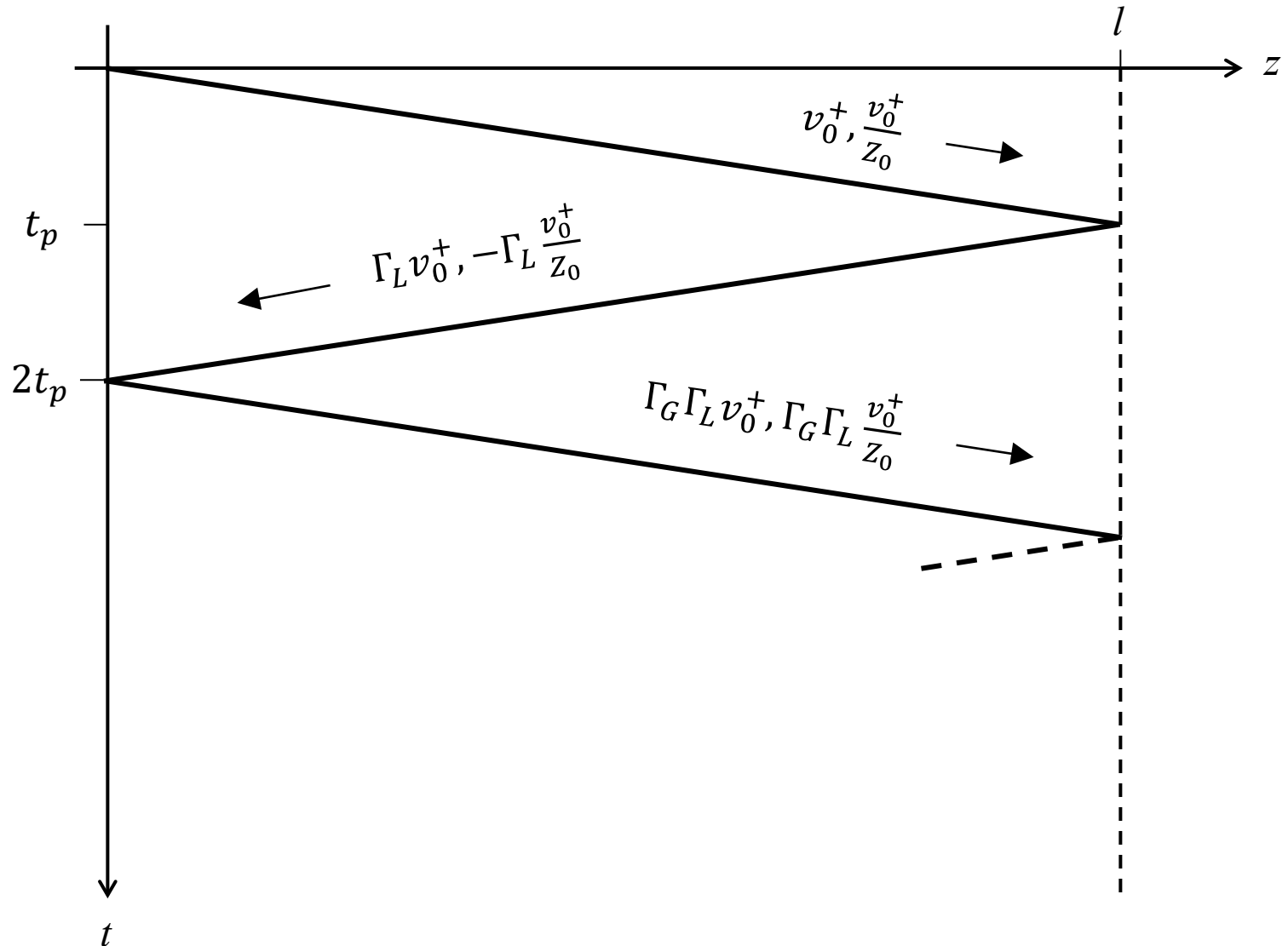
$$\Gamma = \frac{v^-}{v^+} = \frac{R_L - Z_0}{R_L + Z_0}$$

holds also in the time domain.

We consider a line with a generator at one end, and a resistive load R_L at the other. The internal impedance of the generator is real, R_G

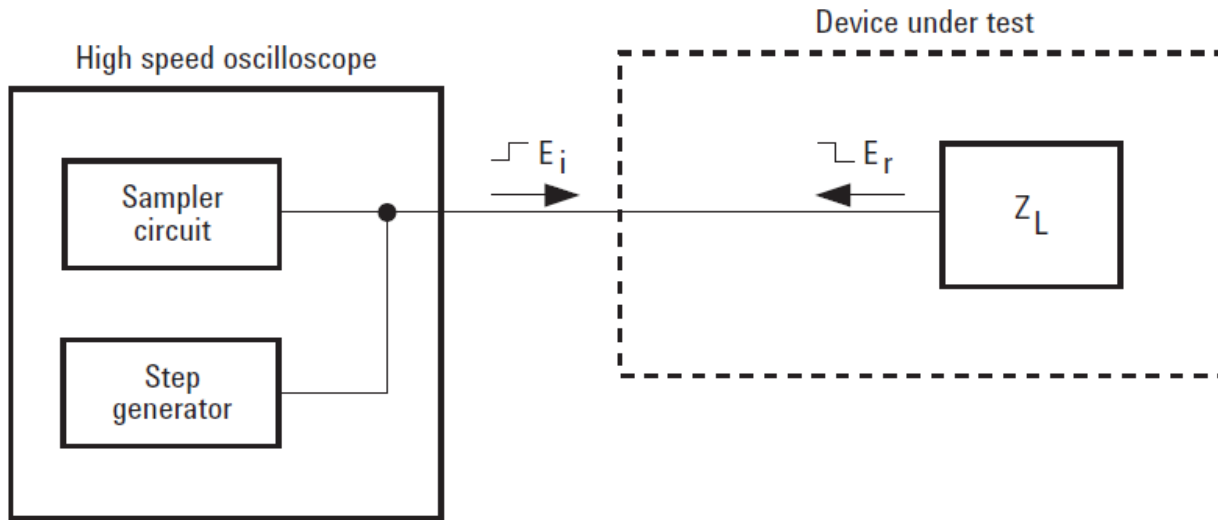


Reflection diagram



Time-domain reflectometry

It is often difficult to locate the exact place where a cable is broken or bent, a connector is loose, etc... However, each of these faults causes a local discontinuity in the characteristic impedance $\Rightarrow$ reflections



Time-domain reflectometry is a method where a step (or pulse) generator sends a pulse down the line, and a fast oscilloscope records the reflected signal. The delay between the pulse and the reflection tells you the distance between the fault and the generator!

Power along a line

Consider a lossless line. Recall Eqns. (18)

$$V(z) = V_0^+ (e^{-j\beta z} + \Gamma e^{+j\beta z}) \quad (18a)$$

$$I(z) = \frac{V_0^+}{Z_0} (e^{-j\beta z} - \Gamma e^{+j\beta z}) \quad (18b)$$

The average power at any point is:

$$P_{av}(z) = \frac{1}{2} \text{Re}\{V(z)I^*(z)\} \quad (19)$$

Recall that these are phasors, i.e. we are considering steady-state sinusoidal signals of the form $v(z, t) = V(z)\cos(\omega t + \varphi)$ and so on. In other words, V is the peak value of the sinusoid. “Average power” means “averaged over one period of the sinusoid”. That’s why there is a factor $\frac{1}{2}$ in Eq. (19).

$$\begin{aligned} P_{av}(z) &= \frac{1}{2} \text{Re} \left\{ \frac{|V_0^+|^2}{Z_0} (e^{-j\beta z} + \Gamma e^{+j\beta z})(e^{+j\beta z} - \Gamma^* e^{-j\beta z}) \right\} = \\ &= \frac{1}{2} \frac{|V_0^+|^2}{Z_0} \text{Re}\{1 - \Gamma^* e^{-j2\beta z} + \Gamma e^{+j2\beta z} - |\Gamma|^2\} \end{aligned}$$

Power along a line

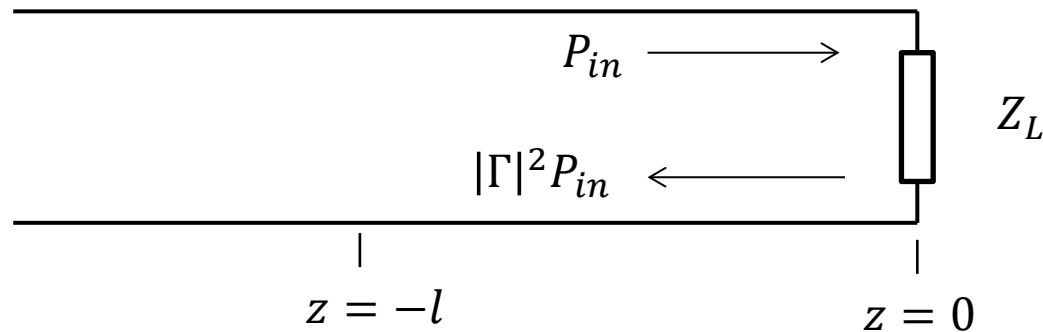
$$P_{av}(z) = \frac{1}{2} \frac{|V_0^+|^2}{Z_0} (1 - |\Gamma|^2) \quad (20)$$

Recall that we are looking at steady-state sinusoidal signals (we used the phasors representation).

Eq. (20) has a simple meaning: the time-averaged power flowing along the line is **constant** (i.e. independent of z) and consists of :

- The **incident** power $\frac{1}{2} \frac{|V_0^+|^2}{Z_0}$ sent towards the load,
- Minus a **reflected** power $\frac{1}{2} |\Gamma|^2 \frac{|V_0^+|^2}{Z_0}$ coming back from the load if it is mismatched, i.e. $\Gamma \neq 0$.

Maximum power at the load



From Eq. (20) follows that the maximum power is obtained when $\Gamma = 0$

⇒ Matching the load to the characteristic impedance of the line has two beneficial effects:

1. Eliminates transients in the time domain
2. Ensures the maximum power is delivered to the load.

Return loss

$$RL = -20 \text{ Log}|\Gamma| \quad (21)$$

The return loss expresses in dB the fraction of power that the line “loses to the load”, i.e. delivers without reflection.

It is often quoted by suppliers of microwave components

$$\begin{array}{lll} |\Gamma| = 0 & \Rightarrow & RL = \infty \quad (\text{all power in the line is delivered to the load}) \\ |\Gamma| = 1 & \Rightarrow & RL = 0 \quad (\text{all power in the line stays on the line, i.e.} \\ & & \text{returns back towards the generator because of} \\ & & \text{reflection}) \end{array}$$

Standing waves

Although the power is constant along the line, the voltage and the current are not necessarily so.

Using Eq. (18a):

$$|V(z)| = |V_0^+ e^{-j\beta z} (1 + \Gamma e^{+j2\beta z})| = |V_0^+| |1 + \Gamma e^{+j2\beta z}|$$

At a point $z = -l$ backwards from the load, and recalling that $\Gamma = |\Gamma|e^{j\theta}$

$$|V(-l)| = |V_0^+| |1 + |\Gamma| e^{j(\theta-2\beta l)}| \quad (22)$$

$|V(-l)|$ is maximum where $e^{j(\theta-2\beta l)} = 1 \Rightarrow l = \frac{\theta}{2\beta}$

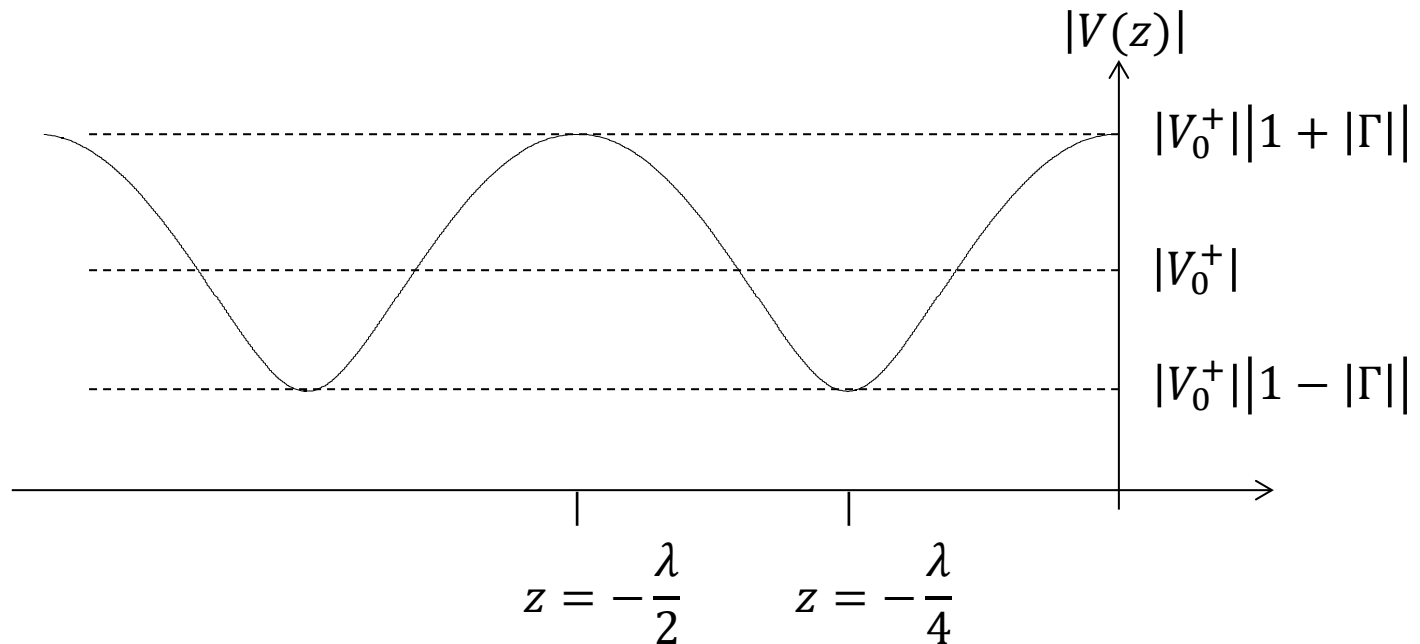
$|V(-l)|$ is minimum where $e^{j(\theta-2\beta l)} = -1 \Rightarrow l = \frac{\theta-\pi}{2\beta}$

A pattern of minima and maxima repeats itself with period $L = \frac{\pi}{\beta}$

→ **Standing waves**

Standing waves

Example: $Z_L = R_L = 150 \, \Omega$; $Z_0 = 50 \, \Omega \Rightarrow \Gamma = \frac{R_L - Z_0}{R_L + Z_0} = \frac{1}{2} = \frac{1}{2} e^{j0}$



Recall Eq. (8): $\lambda = \frac{2\pi}{\beta} \Rightarrow L = \frac{\lambda}{2}$

the standing wave pattern repeats every half-wavelength

Voltage Standing Wave Ratio (VSWR)

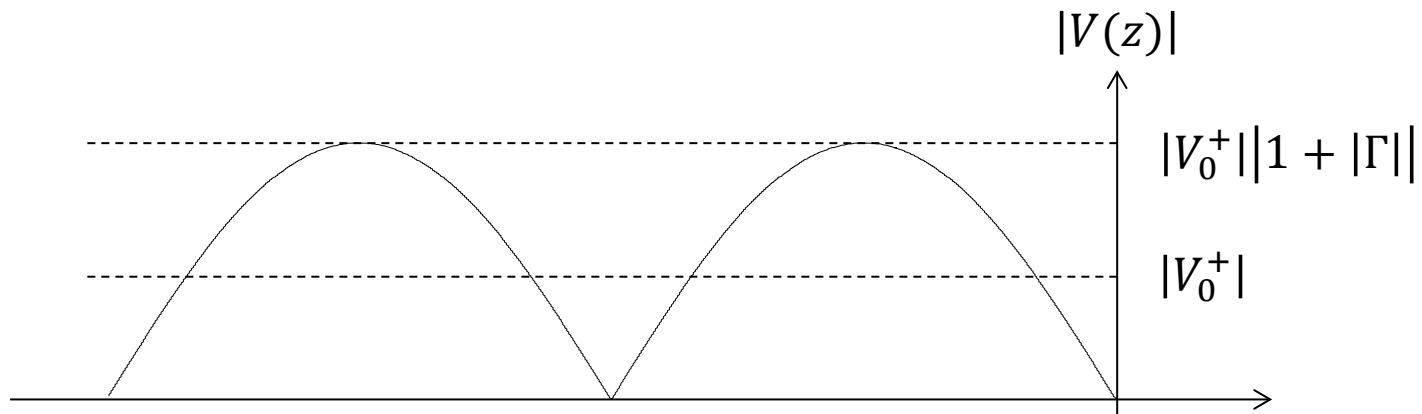
The VSWR is the ratio between the maximum and the minimum voltage in the standing wave pattern:

$$VSWR = S = \frac{V_{max}}{V_{min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad (23)$$

Ideally we want $S = 1$ ($\Gamma = 0$, matched line).

Since $0 \leq |\Gamma| \leq 1$, then $1 \leq S \leq \infty$

Example: short-circuited line, $\Gamma = -1 \Rightarrow S = \infty$



Slotted line



The slotted line is a simple tool to probe the voltage along a line, by sliding a probe through a slot. This allows the extraction of the VSWR, and from there the reflection coefficient, etc...

Input impedance

We learned that:

1. The time-averaged power P_{av} along the line is constant
2. The voltage along the line can have a standing-wave pattern if the load is mismatched

⇒ The current must oscillate with a pattern opposite that of the voltage, to keep P_{av} constant

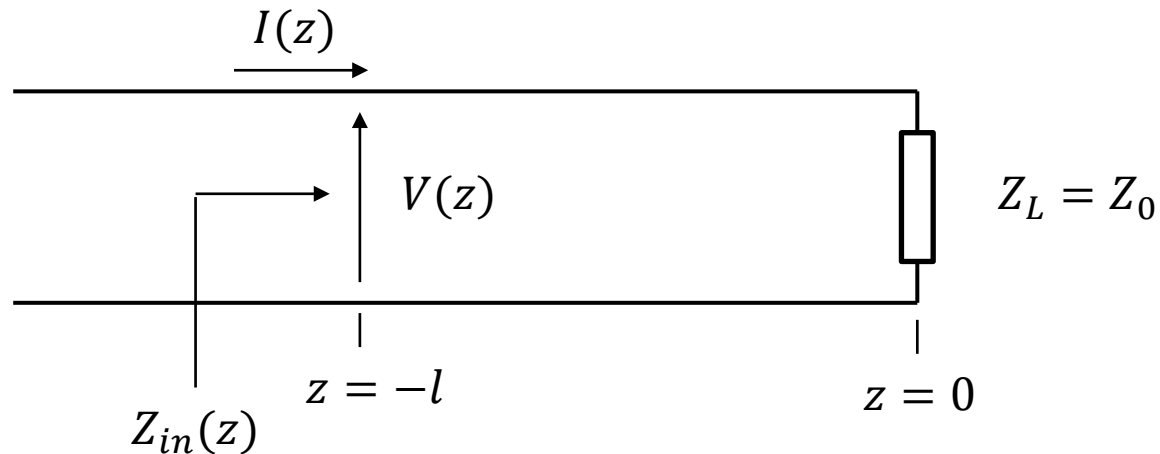
⇒ The impedance seen looking into the line must oscillate accordingly

Input impedance of the line at position $z = -l$, seen looking towards the load:

$$Z_{in} = \frac{V(-l)}{I(-l)} = \frac{V_0^+ (e^{j\beta l} + \Gamma e^{-j\beta l})}{\frac{V_0^+}{Z_0} (e^{j\beta l} - \Gamma e^{-j\beta l})}$$

$$Z_{in} = Z_0 \frac{1 + \Gamma e^{-j2\beta l}}{1 - \Gamma e^{-j2\beta l}} \quad (24)$$

Special case: matched line



$$\Gamma = 0; \quad VSWR = 1$$

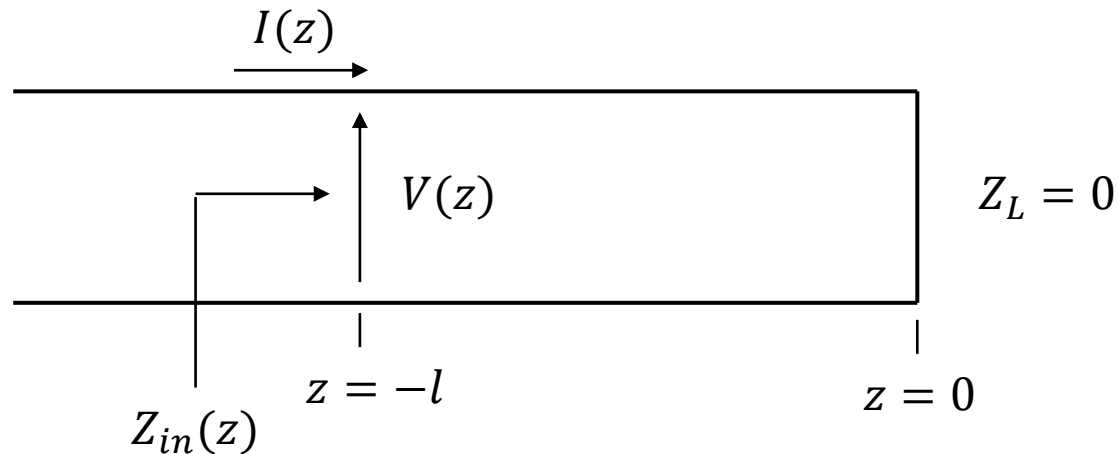
$$V(z) = V_0^+ e^{-j\beta z} \quad \forall z$$

$$I(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} \quad \forall z$$

$$Z_{in}(z) = Z_0 \quad \forall z$$

An impedance-matched line is often called “**flat**”: the voltage, current and impedance are the same no matter where you look along the line.

Special case: shorted line



$$\Gamma = -1; \quad VSWR = \infty$$

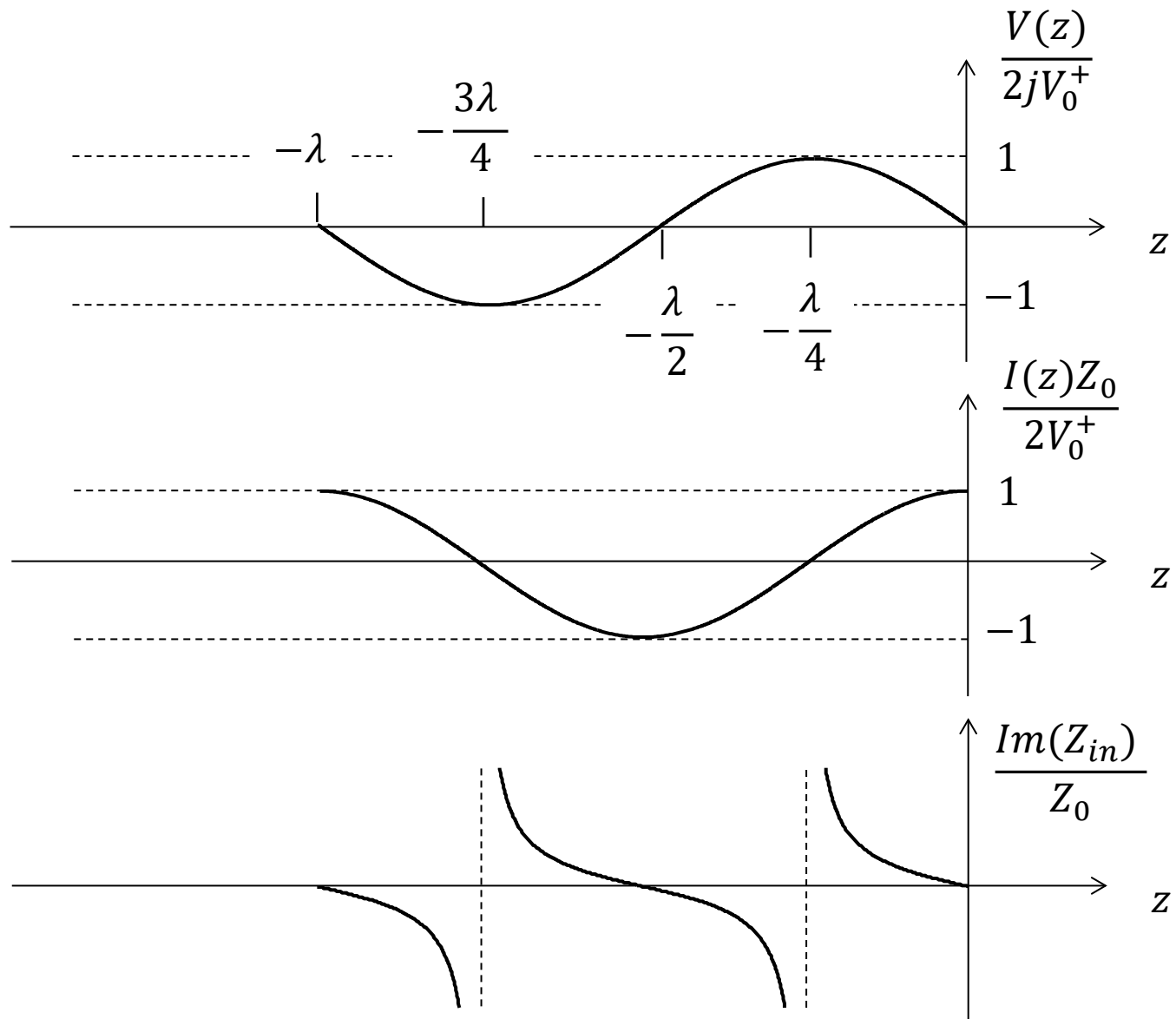
$$V(z) = V_0^+ (e^{-j\beta z} - e^{+j\beta z}) = -2jV_0^+ \sin(\beta z) = 2jV_0^+ \sin(\beta l)$$

$$I(z) = \frac{V_0^+}{Z_0} (e^{-j\beta z} + e^{+j\beta z}) = \frac{2V_0^+}{Z_0} \cos(\beta z) = \frac{2V_0^+}{Z_0} \cos(\beta l)$$

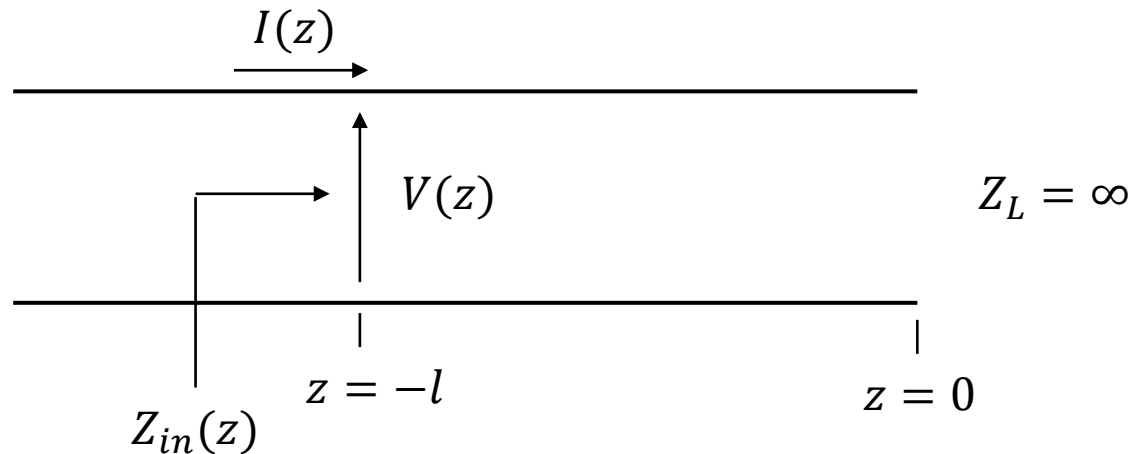
$$Z_{in}^{sc}(z) = -jZ_0 \tan(\beta z) = jZ_0 \tan(\beta l) \quad (25)$$

The input impedance of a shorted line is **purely imaginary**.

Special case: shorted line



Special case: open line



$$\Gamma = 1; \quad VSWR = \infty$$

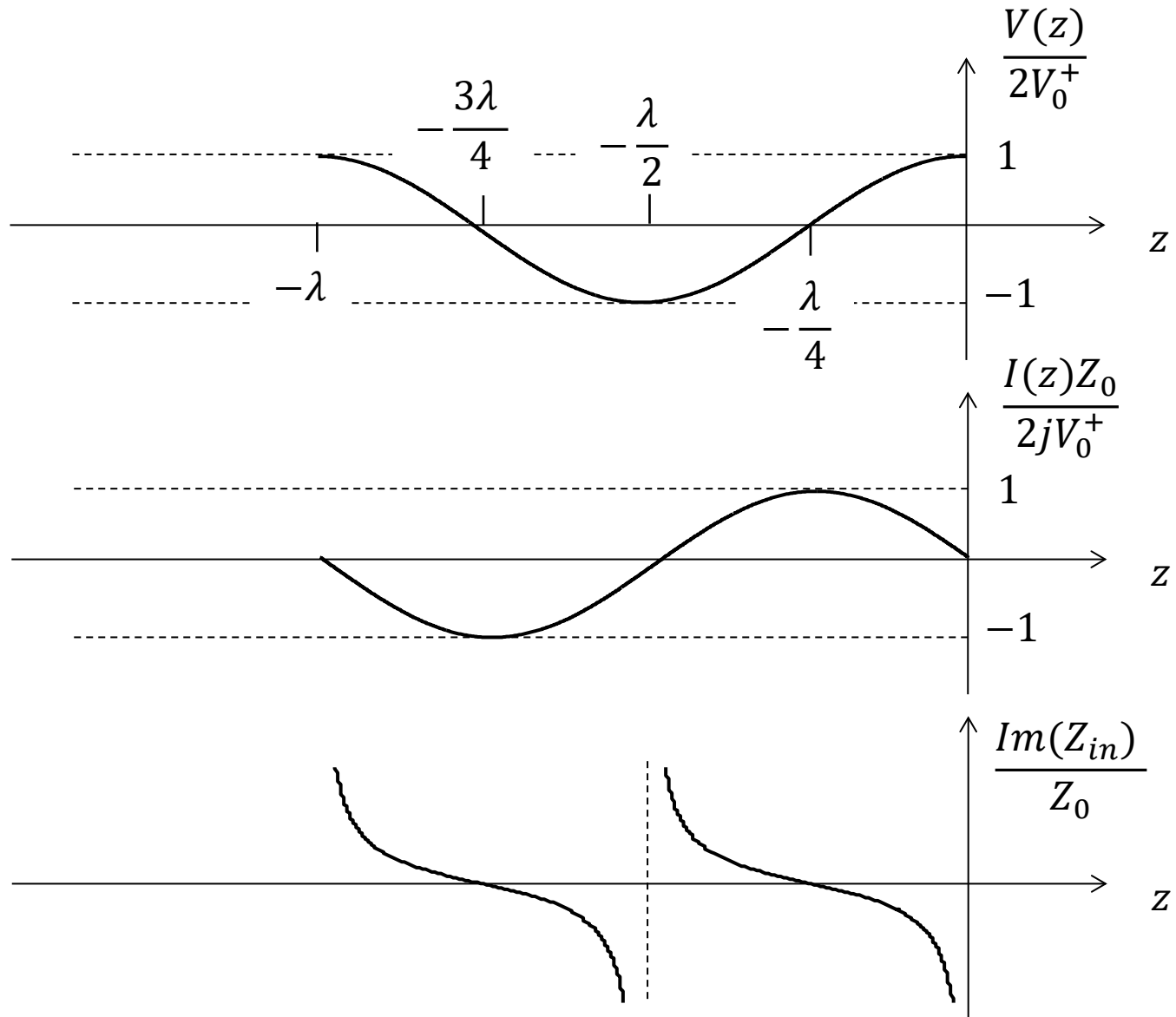
$$V(z) = V(z) = V_0^+ (e^{-j\beta z} + e^{+j\beta z}) = 2V_0^+ \cos(\beta z) = 2V_0^+ \cos(\beta l)$$

$$I(z) = \frac{V_0^+}{Z_0} (e^{-j\beta z} - e^{+j\beta z}) = -j \frac{2V_0^+}{Z_0} \sin(\beta z) = j \frac{2V_0^+}{Z_0} \sin(\beta l)$$

$$Z_{in}^{oc}(z) = jZ_0 \cot(\beta z) = -jZ_0 \cot(\beta l) \quad (26)$$

The input impedance of an open line is **purely imaginary**.

Special case: open line



Line calibration

Shorted and open lines are often used as fixed-point calibrations to extract the characteristic impedance Z_0 and the phase factor β of an unknown line.

Take the product of Eqns. (25) and (26):

$$Z_{in}^{sc}(z)Z_{in}^{oc}(z) = -jZ_0 \tan(\beta z) \times jZ_0 \cot(\beta z) = Z_0^2$$
$$Z_0 = \sqrt{Z_{in}^{sc} Z_{in}^{oc}} \quad (27)$$

Notice that Eq. (27) is independent of z : it doesn't matter where you measure the impedances, as long as you have a short and open at the end of the line.

Taking the ratio:

$$\tan(\beta z) = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}} \quad (28)$$

Here, knowing z you can extract β

Half-wavelength line

In Eq. (24), set $l = \lambda/2$: $e^{\pm j2\beta\lambda/2} = e^{\pm j2\pi} = 1$

$$Z_{in} = Z_0 \frac{1 + \Gamma}{1 - \Gamma} = Z_0 \frac{1 + \frac{Z_L - Z_0}{Z_L + Z_0}}{1 - \frac{Z_L - Z_0}{Z_L + Z_0}} = Z_0 \frac{Z_L + Z_0 + Z_L - Z_0}{Z_L + Z_0 - Z_L + Z_0} = Z_0 \frac{2Z_L}{2Z_0}$$

$$Z_{in} = Z_L$$

The input impedance of a half-wavelength line is the same as the impedance of the load.

It's as if the line weren't there at all!

This holds for any integer multiples of $\lambda/2$

Quarter-wavelength line

In Eq. (24), set $l = \lambda/4$: $e^{\pm j2\beta\lambda/4} = e^{\pm j\pi} = -1$

$$Z_{in} = Z_0 \frac{1 - \Gamma}{1 + \Gamma} = Z_0 \frac{1 - \frac{Z_L - Z_0}{Z_L + Z_0}}{1 + \frac{Z_L - Z_0}{Z_L + Z_0}} = Z_0 \frac{Z_L + Z_0 - Z_L + Z_0}{Z_L + Z_0 + Z_L - Z_0} = Z_0 \frac{2Z_0}{2Z_L}$$

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

The input impedance of a quarter-wavelength line is inversely proportional to the impedance of the load.

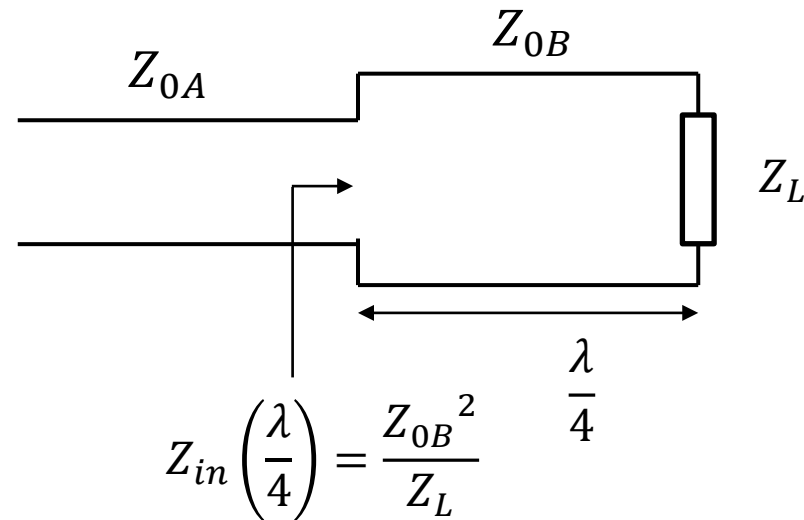
A $\lambda/4$ line transforms an open circuit into a short circuit and vice versa.

This still holds when adding any integer multiples of $\lambda/2$.

Quarter-wave transformer

A $\lambda/4$ line is sometimes used for impedance matching.

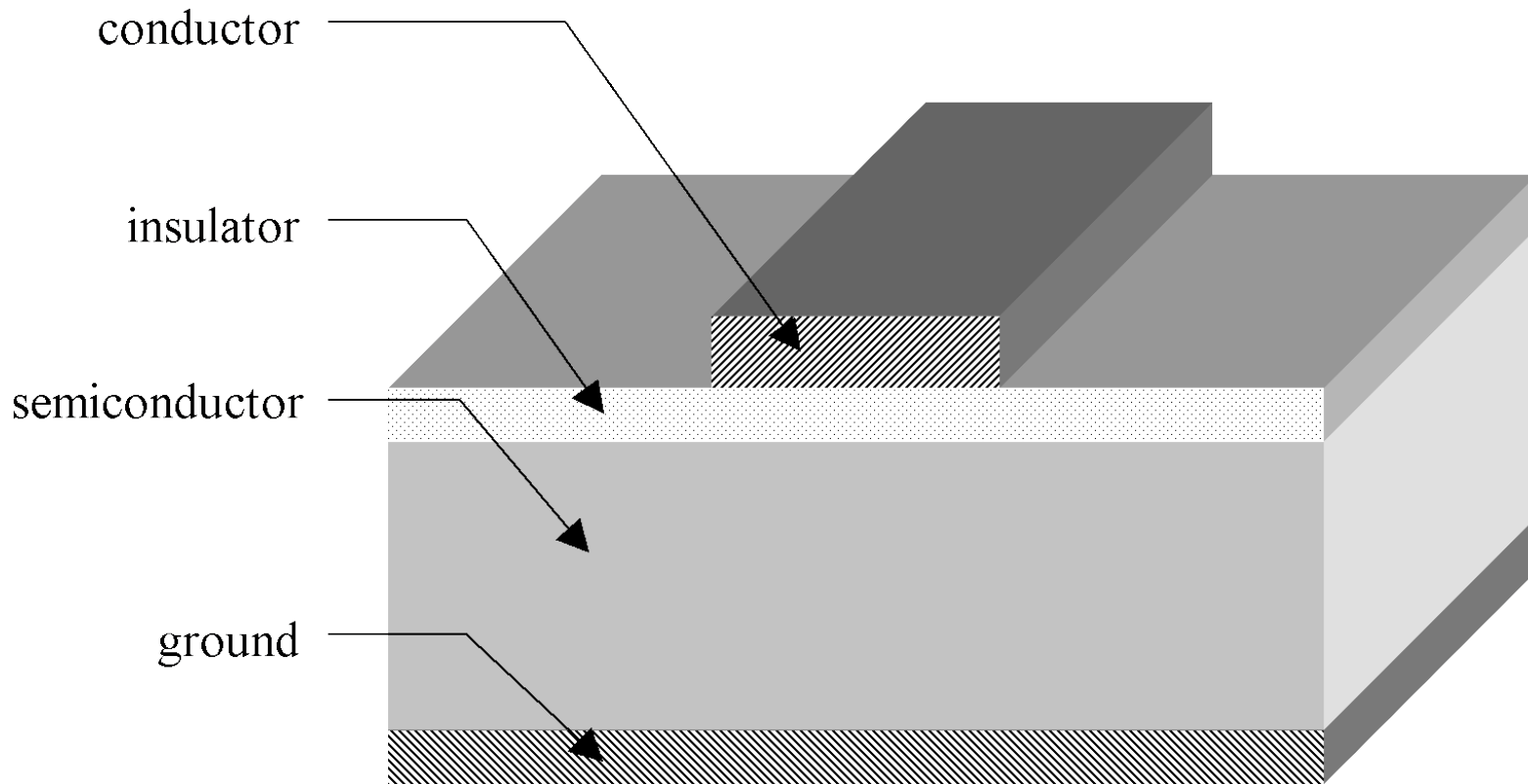
Suppose I have a line with impedance Z_{0A} , a load $Z_L \neq Z_{0A}$, and I can insert a $\lambda/4$ line of impedance Z_{0B} between the two.



If I choose $Z_{0B} = \sqrt{Z_L Z_{0A}}$, then $Z_{in} = Z_{0A}$ and the line is matched.

This technique is useful in stripline circuits, where Z_0 is easily set by the geometry of the strip.

Microstrip



Z_0 is easily changed by varying the width of the upper conductor.